SYED OMER HUSSAIN

Machine Learning Engineer

Chicago | Applied ML | RAG | LLM Evaluation

Governors State University | M.S. Computer Science | Expected May 2027 | GPA 3.90

SELECTED TECHNICAL WORK

RAG for technical learning

Retrieval, generation, and reproducible evaluation

30
public documents

9
retrieval configurations

40
evaluation questions

Built a local assistant with MiniLM embeddings, persistent Chroma storage, and Qwen generation. Selected retrieval settings on validation questions before inspecting the held-out test set.

Python | PyTorch | Sentence Transformers | Chroma | Transformers

LoRA fine-tuning for quantum tutoring

Parameter-efficient adaptation with explicit split control

1,000
instruction examples

800 / 100 / 100
train / validation / test

50
paired evaluation prompts

Adapted Qwen2.5-0.5B-Instruct with completion-only LoRA training. Kept related question families in one split and verified adapter saving, reloading, and base-versus-adapter evaluation.

Python | PyTorch | Hugging Face Transformers | PEFT | Apple MPS

CrossRhythm: cardiac-signal classification

Representation comparison and generalization under dataset shift

24
engineered interval features

LOSO
subject-grouped evaluation

2
datasets: Simband, MIMIC-III

Compared Chronos-Bolt waveform and R-R tachogram embeddings with engineered features for NSR, AF, and PAC classification. Evaluated cross-dataset transfer and CORAL feature alignment.

Python | PyTorch | scikit-learn | Signal Processing | Model Evaluation

My focus Build useful models, make experiments reproducible, and connect results to usable applications.

github.com/iamomer2707

Evaluation evidence

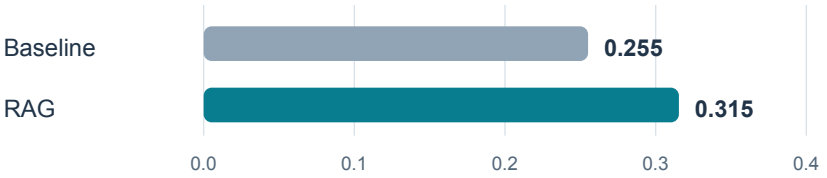
Measured scope, comparison design, and inspectable work

01 RAG: retrieval and generation measured separately

Qwen2.5-0.5B-Instruct | MiniLM embeddings | top-k = 3

Answer token F1

Lexical overlap on 22 answerable test questions



21 / 22

answerable test questions with a relevant source retrieved

Selection: 12 validation questions. Final test: 28 questions, including 6 unanswerable.

Small evaluation set; token F1 measures lexical overlap. Human factuality review remains pending. Sources were exposed through retrieval metadata; valid inline citations were not produced in the measured run.

02 LoRA: disciplined training and comparison

1,000 examples across 130 groups; no group overlap between splits



800 training | 100 val | 100 test

Completion-only training on Apple MPS; paired base/adapter responses on 50 prompts. Saved raw outputs, training configuration, and reload checks. Completion loss and response metrics were recorded as separate diagnostics.

03 CrossRhythm: evaluating dataset shift

Used leave-one-subject-out evaluation within Simband and transfer to MIMIC-III. Reported balanced accuracy, macro-F1, and class-level precision/recall; compared features and CORAL alignment.

Exploratory small-cohort biosignal study. Reference beat annotations were used where available; the results do not establish clinical performance.

PUBLIC WORK TO INSPECT

Speech Enhancement Toolkit - PyTorch, STFT, FastAPI

MeterTrace - live browser application

Evidence basis: completed local experiment reports, retained evaluation outputs, and source code. Portfolio projects are distinct from peer-reviewed publications.